Introduction

Magnetic field at sensor: Tool to measure the valid air-gap (distances from magnet surface to sensor) given a certain magnet size and remanence (diametrical magnetization). Worst case angle error due to assembly tolerances: Tool to measure the worst case angle error caused by tilts and eccentricites of magnet and sensor elements against the axis of rotation. The worst case angle error is taken at the worst rotational position for the maximum assembly tolerances combination. Simulation results based on worst case scenario. NOTE: Tool describes purely the error contribution due to assembly tolerances. Additional tolerances based on the ICs temperature and life time drift are not considered!Please refer to datasheet, www.infineon.com/sensors

Disclaimer

The intention of the Infineon Solution Finder is to offer a quick product and application search covering the whole Infineon product- and application portfolio. For improvement and maintenance reasons all data & algorithms are subject to change without prior notice. Calculations & simulations are based on simplified assumptions and operating condition which may not reflect the complexity of your real world system. For these reasons Infineon cannot take any responsibility or liability for the results generated by the Solution Finder's program. Please take this into account and carefully study all datasheet & application notes when designing your system. For questions please contact our support team!

Specification

Magnet Properties

Magnet Remanence:	1200 mT
Magnet Thickness:	2.5 mm
Magnet Diameter:	6 mm

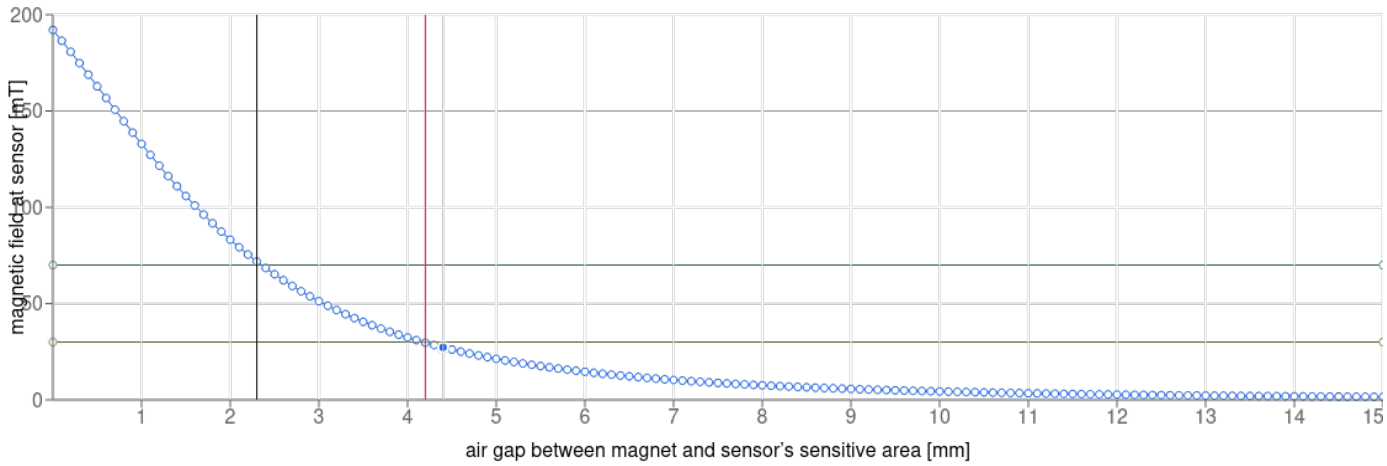
Sensor Specification

Bmin required:	30 mT, @Tambient
Bmax required:	70 mT, @Tambient

Assembly Tolerances

Air Gap:	1 mm
Maximum Beta:	2 degree
Maximum Lambda:	3 degree
Maximum Delta:	200 * 10^-6 m
Maximum Epsilon:	300 * 10^-6 m

Magnet Field at Sensor (with non magnetic shafts)



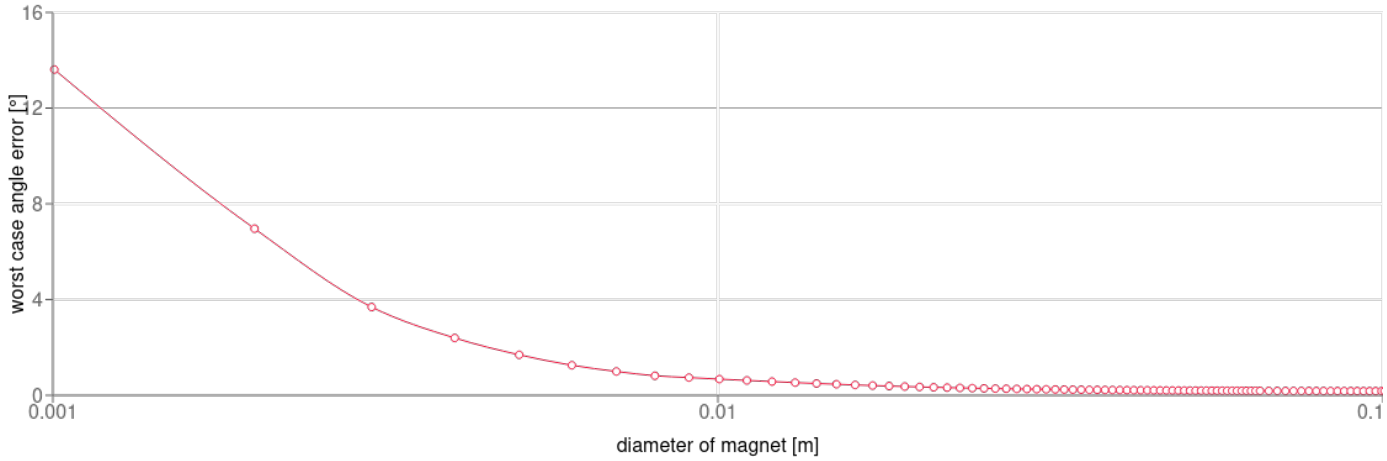
min air gap [mm]:	2.3	max air gap [mm]:	4.2

air gap between magnet and sensor's sensitive area [mm]	magnetic field at sensor [mT]
0	192.055
0.1	186.485
0.2	180.734
0.3	174.844
0.4	168.858
0.5	162.812
0.6	156.746
0.7	150.692
0.8	144.683
0.9	138.747
1	132.909
1.1	127.190
1.2	121.610
1.3	116.183
1.4	110.922
1.5	105.836
1.6	100.933
1.7	96.216
1.8	91.689
1.9	87.353
2	83.205
2.1	79.244
2.2	75.468
2.3	71.870
2.4	68.447
2.5	65.193
2.6	62.101
2.7	59.166
2.8	56.382
2.9	53.741
3	51.237
3.1	48.863
3.2	46.614
3.3	44.483
3.4	42.464
3.5	40.551
3.6	38.739
3.7	37.021
3.8	35.393
3.9	33.850
4	32.388
4.1	31.001
4.2	29.685
4.3	28.437
4.4	27.253
4.5	26.128
4.6	25.061
4.7	24.046
4.8	23.083
4.9	22.166

air gap between magnet and sensor's sensitive area [mm]	magnetic field at sensor [mT]
5	21.295
5.1	20.466
5.2	19.678
5.3	18.927
5.4	18.211
5.5	17.530
5.6	16.881
5.7	16.261
5.8	15.671
5.9	15.107
6	14.569
6.1	14.055
6.2	13.564
6.3	13.095
6.4	12.646
6.5	12.217
6.6	11.807
6.7	11.414
6.8	11.037
6.9	10.677
7	10.331
7.1	10.000
7.2	9.683
7.3	9.378
7.4	9.086
7.5	8.805
7.6	8.535
7.7	8.276
7.8	8.028
7.9	7.788
8	7.558
8.1	7.337
8.2	7.124
8.3	6.919
8.4	6.722
8.5	6.532
8.6	6.349
8.7	6.173
8.8	6.002
8.9	5.838
9	5.680
9.1	5.528
9.2	5.380
9.3	5.238
9.4	5.101
9.5	4.968
9.6	4.840
9.7	4.716
9.8	4.596
9.9	4.480

air gap between magnet and sensor [mm]	magnetic field at sensor [mT]
10	4.368
10.1	4.260
10.2	4.155
10.3	4.054
10.4	3.955
10.5	3.860
10.6	3.768
10.7	3.679
10.8	3.592
10.9	3.508
11	3.427
11.1	3.348
11.2	3.272
11.3	3.197
11.4	3.125
11.5	3.056
11.6	2.988
11.7	2.922
11.8	2.858
11.9	2.796
12	2.735
12.1	2.677
12.2	2.620
12.3	2.564
12.4	2.510
12.5	2.458
12.6	2.407
12.7	2.357
12.8	2.309
12.9	2.262
13	2.217
13.1	2.172
13.2	2.129
13.3	2.087
13.4	2.046
13.5	2.006
13.6	1.967
13.7	1.929
13.8	1.892
13.9	1.855
14	1.820
14.1	1.786
14.2	1.752
14.3	1.720
14.4	1.688
14.5	1.657
14.6	1.627
14.7	1.597
14.8	1.568
14.9	1.540
15	1.512

Worst Case Angle Error due to Assembly Tolerances



diameter of magnet [m]	worst case angle error [degree]
0.001	13.612
0.002	6.962
0.003	3.685
0.004	2.393
0.005	1.690
0.006	1.253
0.007	0.994
0.008	0.814
0.009	0.738
0.01	0.673
0.011	0.619
0.012	0.571
0.013	0.528
0.014	0.491
0.015	0.459
0.016	0.430
0.017	0.405
0.018	0.384
0.019	0.364
0.02	0.348
0.021	0.332
0.022	0.319
0.023	0.307
0.024	0.297
0.025	0.287
0.026	0.278
0.027	0.271

diameter of magnet [m]	worst case angle error [degree]
0.028	0.264
0.029	0.257
0.03	0.251
0.031	0.246
0.032	0.241
0.033	0.237
0.034	0.232
0.035	0.229
0.036	0.225
0.037	0.222
0.038	0.219
0.039	0.216
0.04	0.214
0.041	0.211
0.042	0.209
0.043	0.207
0.044	0.205
0.045	0.203
0.046	0.201
0.047	0.200
0.048	0.198
0.049	0.197
0.05	0.195
0.051	0.194
0.052	0.193
0.053	0.192
0.054	0.191

diameter of magnet [m]	worst case angle error [degree]
0.055	0.190
0.056	0.189
0.057	0.188
0.058	0.187
0.059	0.186
0.06	0.185
0.061	0.184
0.062	0.184
0.063	0.183
0.064	0.182
0.065	0.182
0.067	0.180
0.069	0.179
0.071	0.178
0.073	0.178
0.075	0.177
0.077	0.176
0.079	0.175
0.081	0.175
0.083	0.174
0.085	0.173
0.087	0.173
0.089	0.172
0.091	0.172
0.093	0.171
0.095	0.171
0.097	0.171
0.099	0.170
0.1	0.170